

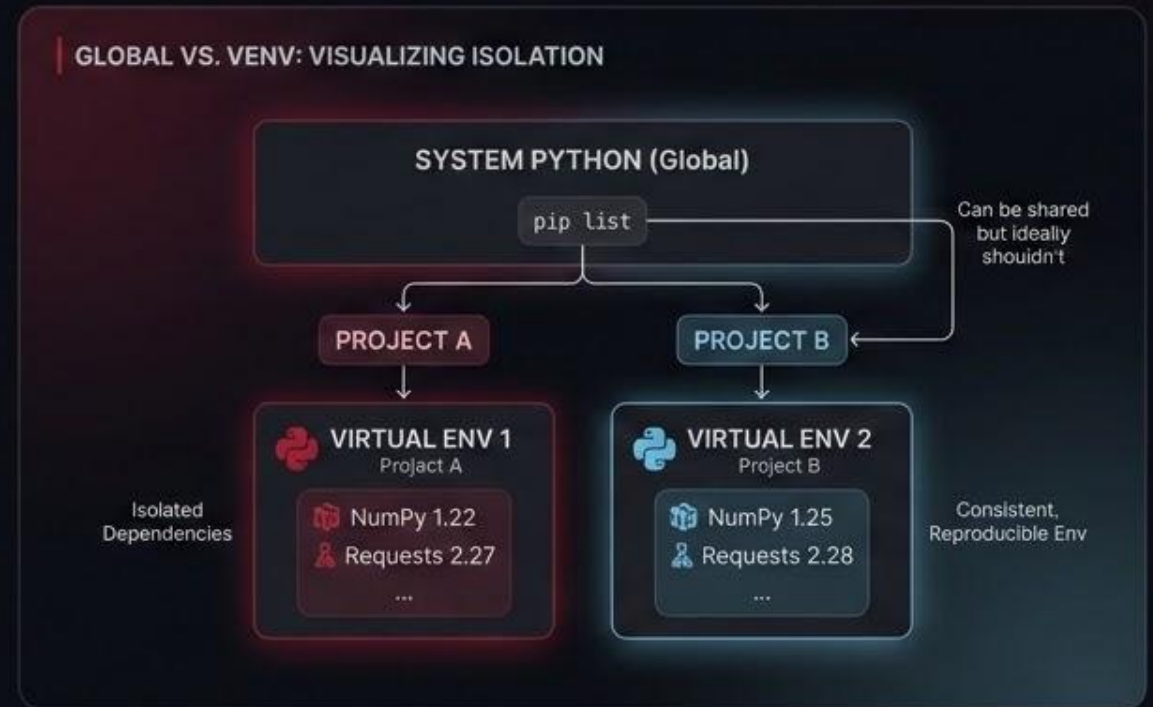
PYTHON BASICS

Week 1: Environment, Variables, & Control Flow

MIT KATHMANDU — ACCELERATED COURSE

| WHY VIRTUAL ENVIRONMENTS?

- 🛡️ **Isolation:** Prevents package version conflicts between projects.
- 📁 **Project-Specific:** Keep dependencies clean and relevant.
- 🌿 **Clean Installation:** Never clutter your system's global Python.
- 🔗 **Reproducibility:** Easily share exact requirements.



| SETUP: VENV COMMANDS

1. Creation

```
python -m venv venv
```

This creates a folder named venv containing a copy of Python.

2. Activation

Windows:

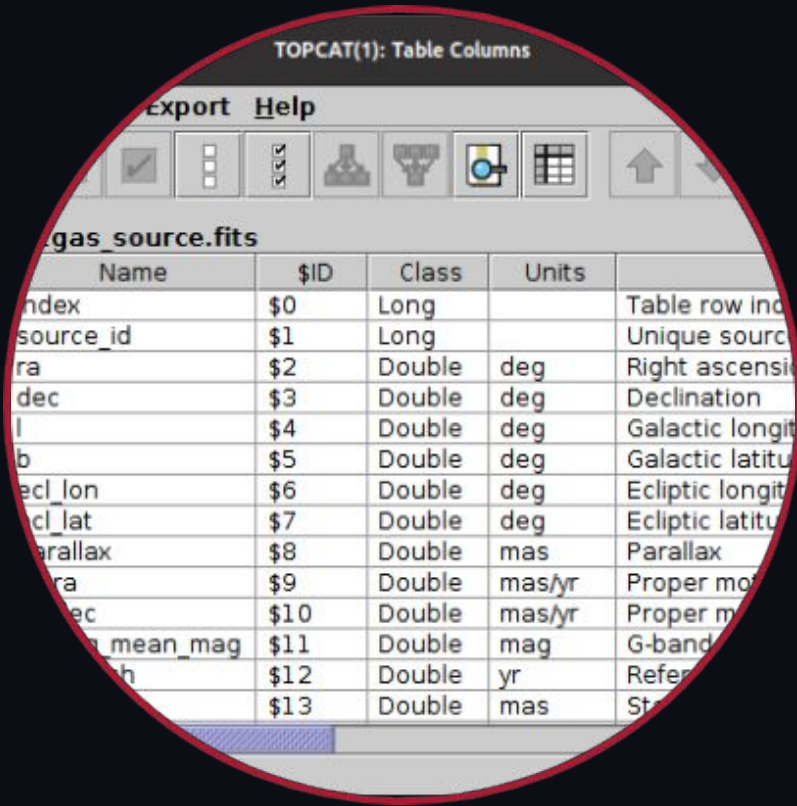
```
.\venv\Scripts\activate
```

macOS / Linux:

```
source venv/bin/activate
```

Note: Your terminal prompt will change to show (venv) at the start.

VARIABLES: DATA CONTAINERS



The screenshot shows the TOPCAT(1): Table Columns window. It features a menu bar with 'Export' and 'Help', and a toolbar with icons for selection, sorting, and other table operations. Below the toolbar is a table with columns: Name, \$ID, Class, Units, and a description. The table lists various astronomical parameters and their units.

Name	\$ID	Class	Units	
index	\$0	Long		Table row index
source_id	\$1	Long		Unique source ID
ra	\$2	Double	deg	Right ascension
dec	\$3	Double	deg	Declination
l	\$4	Double	deg	Galactic longitude
b	\$5	Double	deg	Galactic latitude
ecl_lon	\$6	Double	deg	Ecliptic longitude
ecl_lat	\$7	Double	deg	Ecliptic latitude
parallax	\$8	Double	mas	Parallax
mu_ra	\$9	Double	mas/yr	Proper motion in RA
mu_dec	\$10	Double	mas/yr	Proper motion in Dec
mean_mag	\$11	Double	mag	G-band mean magnitude
h	\$12	Double	yr	Reference epoch
	\$13	Double	mas	Standard error

Naming: Use snake_case (e.g., user_age).

Dynamic: No need to declare type (e.g., int, str).

Assignment: The = operator stores data.

```
name = "Alice"  
age = 25  
is_student = True
```

| CORE DATA TYPES



int

Whole numbers

42, -7, 0



float

Decimal numbers

3.14, -0.5



str

Text sequences

"Hello", 'MIT'



bool

Logic states

True, False

| TYPE CASTING

Explicit Conversion

```
int("42")      # 42
float("3.1")   # 3.1
str(100)       # "100"
```

The input() Pitfall

input() **always** returns a string.

```
age = input("Age? ")
age = int(age) # Must cast!
```

| ARITHMETIC OPERATORS

Operator	Meaning	Example	Result
<code>+, -, *</code>	Add, Sub, Mult	<code>10 * 2</code>	20
<code>/</code>	Division	<code>10 / 4</code>	2.5 (float)
<code>//</code>	Floor Division	<code>10 // 4</code>	2 (int)
<code>%</code>	Modulo (Remainder)	<code>10 % 3</code>	1
<code>**</code>	Exponent	<code>2 ** 3</code>	8

Follows **PEMDAS** precedence rules.

COMPARISON OPERATORS

Symbol	Description	Example	Boolean Result
==	Equal to	5 == 5	True
!=	Not equal to	5 != 3	True
>, <	Greater / Less	10 > 20	False
>=, <=	Greater/Less or Equal	18 >= 18	True

CRITICAL: = is for assignment. == is for comparison!

| LOGICAL OPERATORS



and

True if **both** are True.



or

True if **at least one** is True.



not

Reverses the boolean value.

```
can_drive = (age ≥ 18) and (has_license = True)
```

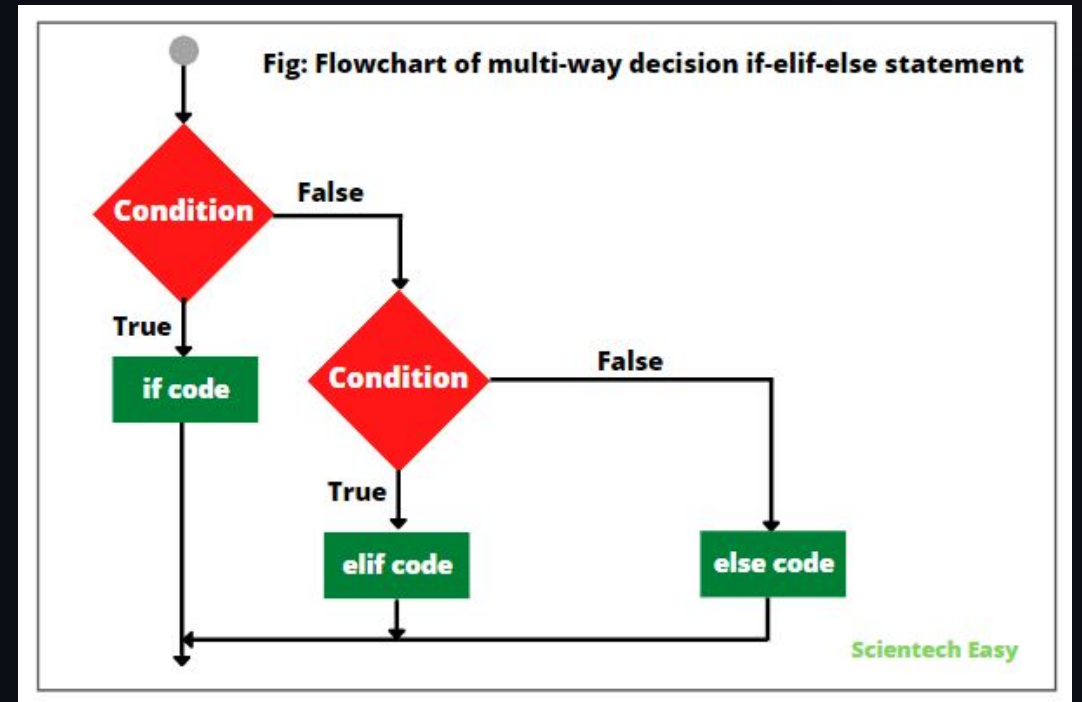
| IF/ELIF/ELSE: DECISIONS

Colons : start the block.

Indentation (4 spaces) defines the body.

elif checks only if previous failed.

```
if score ≥ 90:  
    print("Grade A")  
elif score ≥ 80:  
    print("Grade B")  
else:  
    print("Keep trying!")
```



| TRUTHY & FALSY

Falsy (The "Empty" Set)

- 0, 0.0 (Numeric Zero)
- "" (Empty String)
- None (Empty Object)
- [], {} (Empty Containers)

Truthy

Everything else is considered True!

```
name = "Alice"  
if name: # Shortcut for name != ""  
    print("Hello " + name)
```

| WHILE LOOP

10

Countdown Start

Conditional Repetition

Runs as long as the condition remains True.

```
count = 10
while count > 0:
    print(count)
    count -= 1 # Crucial update!
```

Beware: Infinite loops happen if condition never becomes False.

| FOR LOOP & RANGE()

Sequence Iteration

More readable for known counts.

```
for i in range(5):  
    print(i) # 0, 1, 2, 3, 4
```

The range() Function

- range(stop): Starts at 0.
- range(start, stop): Custom start.
- range(start, stop, step): Skips numbers.

```
range(0, 10, 2) # 0, 2, 4, 6, 8
```

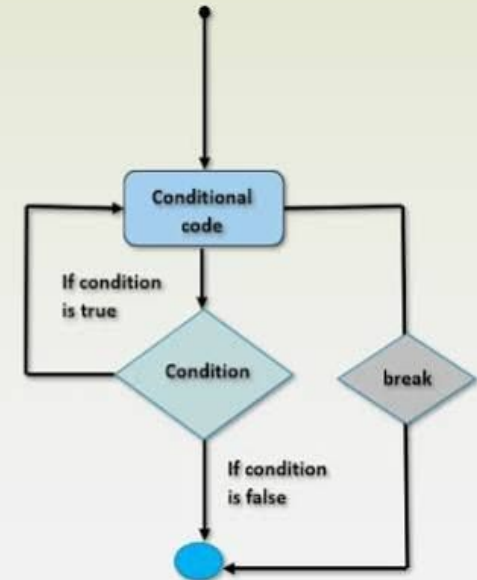
| BREAK: EXIT EARLY

STOP

Immediately kills the innermost loop.

```
for n in range(100):  
    if n == 7:  
        break # Stops at 7  
    print(n)
```

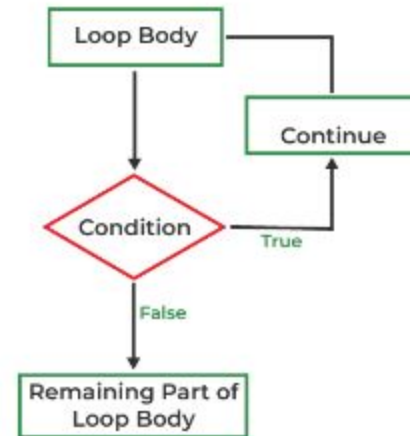
Break Statement in Python



| CONTINUE: SKIP AHEAD

Skips the rest of current iteration and jumps to the next item.

```
for n in range(5):  
    if n == 2:  
        continue  
    print(n) # Prints: 0, 1, 3, 4
```



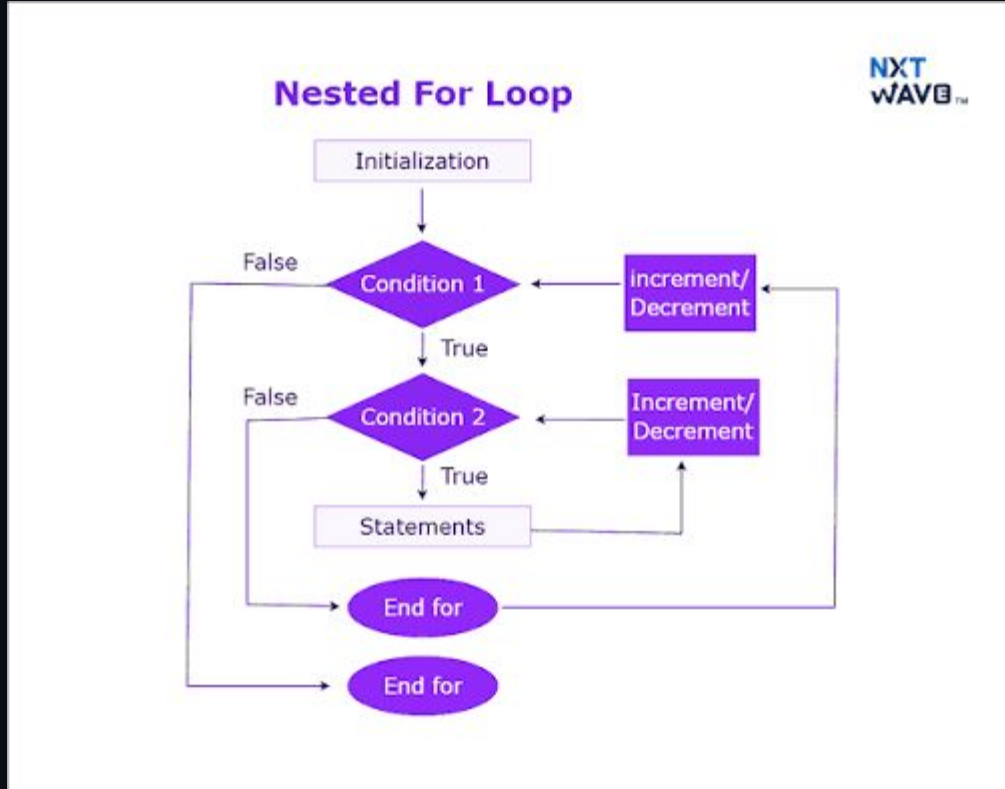
“

pass

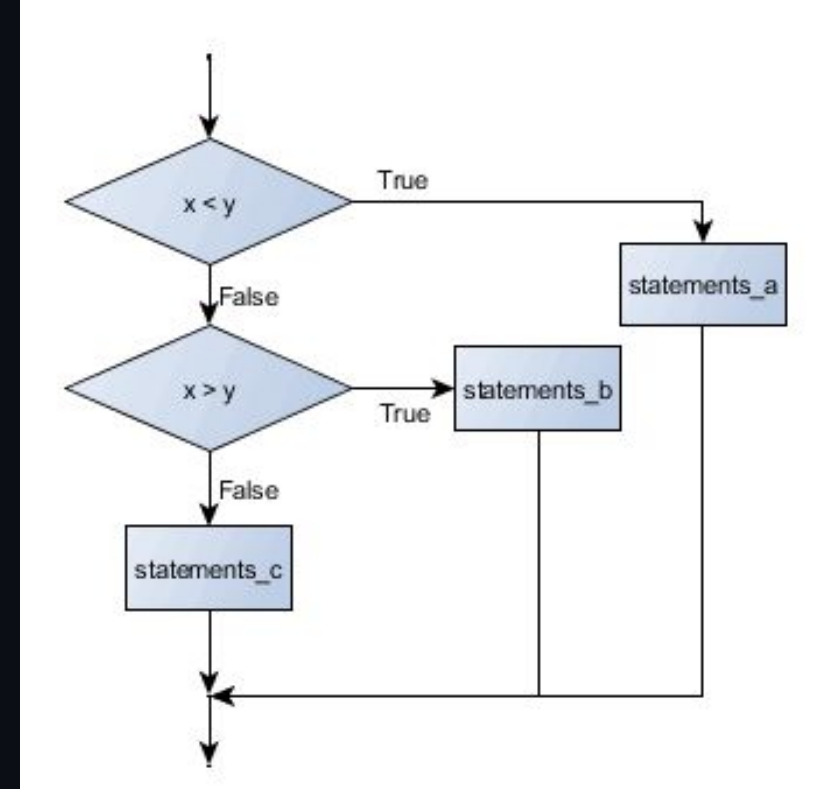
"A placeholder for code you haven't written yet."

```
if user_is_logged_in:  
    pass # TODO: handle logic
```


| NESTED CONTROL FLOW



Loops inside loops.



Conditional branching logic.

DEMO: CONVERTER

```
celsius = float(input("Enter temp (C): " ))  
fahrenheit = (celsius * 9/5) + 32  
  
print(f"{celsius}°C is {fahrenheit}°F")
```

- Input → Casting (float)
- Arithmetic calculation
- f-string formatting

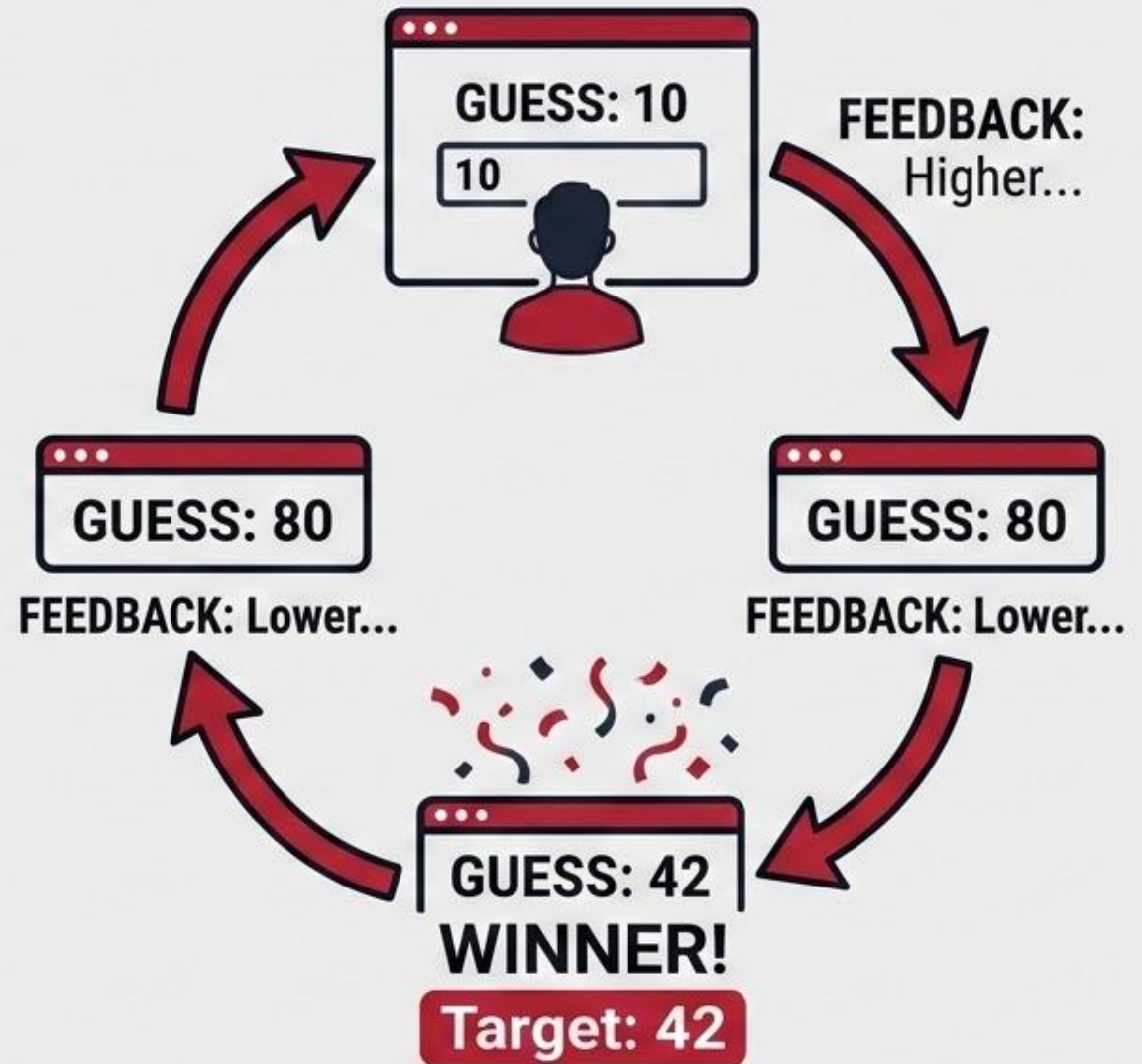


DEMO: GAME

```
secret = 42 while True: guess = int(input("Guess (1-100): ")) if guess == secret: print("You won!") break elif guess < secret: print("Higher...") else: print("Lower...")
```

```
secret = 42
while True:
    guess = int(input("Guess (1-100): "))
    if guess == secret:
        print("You won!")
        break
    elif guess < secret:
        print("Higher...")
    else:
        print("Lower...")
```

- secret = 42
- while True
- guess = input
- if/elif/else
 - print 'Higher...'
 - print 'You won!' break



| RECAP & NEXT STEPS

Venv & Vars

Setup and basic storage

Types & Ops

Math and logic

Control Flow

If, Loops, Break

Weekend Project

ATM Simulator



Next Week: **Data Structures (Lists, Dicts, Tuples)**

| IMAGE SOURCES



<https://preview.redd.it/rant-after-going-through-literal-hell-with-pythons-package-v0-2o8wlapqk5h31.png?width=492&format=png&auto=webp&s=5c57e4c12d1bd808e7239e622c9d442d5892e2fe>

Source: www.reddit.com



<https://www.star.bris.ac.uk/~mbt/topcat/figures/ColumnInfoWindow.png>

Source: www.star.bris.ac.uk



https://lh6.googleusercontent.com/RP8WkNnhwJwIm4j1a_13_ilnw9bCk6x1O4PPJNovFYrneeljilrteGqm_bpMg55LaGEAKLSkXK3UHUthgDcCYUxw5OTdufJr7HmA8-icILBTsC5Bqzu4LixKATQQxwSnq3li6_qt=s0

Source: www.datasource.ai



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<https://files.realpython.com/media/default-vs-custom-syntax-highlighting-python-314.274b304a912a.png>

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IMAGE SOURCES



<https://education.vex.com/stemlabs/sites/default/files/inline-images/CleanShot%202024-11-19%20at%2010.23.33%402x.png>

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